

Supplementary Material

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A RL Teacher Training and Safety Details

A.1 Asymmetric Teacher Observations

Table 1 contrasts the actor’s noisy, deployable signals with the centralized critic’s clean, privileged simulator state. The palm and five fingertips constitute the six hand links. A “twist” is the concatenation of three-dimensional linear and angular velocity. Running mean/variance normalization is applied to both inputs and to value targets.

Table 1: Asymmetric teacher observations.

Signal	Actor	Critic	Definition
Robot joint position q_t	13 (noisy)	13	Actuated arm–hand joint angles (rad)
Robot joint velocity $\dot{q}_t$	13 (noisy)	13	Actuated arm–hand joint velocities (rad/s)
Hand-link position	18 (noisy)	18	Palm and five fingertip positions: 6×3 Cartesian coordinates
Hand-link velocity	18 (noisy)	36	Actor: linear velocity (6×3); critic: spatial twist (6×6)
Palm contact force	–	3	Privileged Cartesian contact-force vector (N)
Measured joint torque	–	19	Privileged torque measurements for 19 instrumented joints (N m)
Object position	3 (noisy)	3	Object Cartesian position
Object orientation	4 (noisy)	4	Object rotation as a unit quaternion
Object velocity	–	6	Privileged object linear and angular velocity (twist)
Goal position	3	3	Cartesian target position for the lift task
Object scale	1	1	Scalar geometry scale supplied by the environment
Object ID	1	1	Scalar identifier for the current object
Previous action a_{t-1}	13	13	Previous 13-dimensional arm–hand command
Total dimension	87	133	

“–” denotes a signal that is unavailable to the actor. Noisy entries are sensor-like estimates; unmarked actor entries are task context or action history. Critic-only entries are used during simulation training and are not available to the deployed policy.

A.2 Termination Conditions

All conditions are checked at 60 Hz in the environment frame; the table-top height is $z_T = 0.25$ m.

- (a) *Harmful collision*: We define a safety-clearance region above the table and monitor all arm–table contacts. We trigger this termination when the palm or fingertips enter the clearance region, or when any arm link contacts the table.
- (b) *Object escape*: We define a permissible object workspace above the table that covers the intended manipulation region. We trigger object-escape termination as soon as the object center leaves this space or falls below the table.
- (c) *Palm flipped*: The palm should face downward toward the object and table. We trigger this termination when its downward direction differs from world-down by more than 90° , at which point the nominal grasp geometry is inverted.
- (d) *Hand too far*: We use the same permissible workspace as in (b) and trigger this termination when the palm link leaves it, indicating that the hand has moved too far during exploration.
- (e) *Timeout*: We terminate the episode when the time index reaches the maximum episode horizon, $t \geq T_{\max}$.

Expanding the termination rule used in the main paper gives

$$\mathbb{I}_{\text{term}}(s_t, t) = \mathbb{I}_{\text{coll}}(s_t) \vee \mathbb{I}_{\text{obj}}(s_t) \vee \mathbb{I}_{\text{flip}}(s_t) \vee \mathbb{I}_{\text{hf}}(s_t) \vee \mathbb{I}_{\text{timeout}}.$$

To avoid conflicts between simultaneous termination conditions, we assign the event label using the following descending severity priority: harmful collision $\succ$ object escape $\succ$ palm flipped $\succ$ hand too far $\succ$ timeout.

A.3 Runs and Random Seeds

Teacher performance in Main Paper Table II is evaluated over 200 episodes per object, and the real-world results are evaluated over 30 episodes per object. Each of the 40 object-specific teacher policies is trained with five independent random seeds $\{1, 2, 3, 4, 5\}$, with the same evaluation protocol applied to every seed. For each seed, teacher performance is evaluated on all 40 objects. The table below reports the mean $\pm$ standard deviation across the 40 object-level rates for lift success, overall unsafe events, and the four unsafe-event categories separately for each seed.

Table 2: Teacher policy performance across all 40 objects for each random seed (200 evaluation episodes per object and seed). Each entry is the mean $\pm$ standard deviation across the 40 object-level rates. Numerical entries are provisional synthetic placeholders pending verification with the measured evaluation data.

Metric (%)	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5
Lift success	86.7 ± 7.8	87.4 ± 7.5	86.9 ± 8.0	87.8 ± 7.3	87.1 ± 7.7
Unsafe rate avg.	14.2 ± 6.1	13.6 ± 5.8	14.0 ± 6.3	13.2 ± 5.6	13.8 ± 6.0
Harmful collision	3.1 ± 3.5	2.9 ± 3.3	3.2 ± 3.6	2.8 ± 3.2	3.0 ± 3.4
Object escape	9.0 ± 5.2	8.8 ± 5.0	8.9 ± 5.3	8.7 ± 4.9	8.9 ± 5.1
Palm flipped	1.2 ± 2.4	1.1 ± 2.3	1.1 ± 2.3	1.0 ± 2.2	1.1 ± 2.3
Hand too far	0.9 ± 1.9	0.8 ± 1.8	0.8 ± 1.8	0.7 ± 1.7	0.8 ± 1.8

B VLM Prompt and Inference Details

B.1 Visual-Input Storage and Example

Candidate visual inputs are retained during a run in a bounded in-memory buffer of at most 64 samples. At each query, up to eight selected samples are embedded as JPEG data URLs in a self-contained `<run_dir>/vlm_reports/*.html` report together with their metadata, the prompt, model reply, parsed JSON, and API diagnostics. The accompanying `vlm_threshold_advisor.jsonl` strips the base64 payloads and retains only metadata; no separate image files are written.

Figure 1 shows four actual inputs extracted from the first archived query report, one for each object in that rollout.

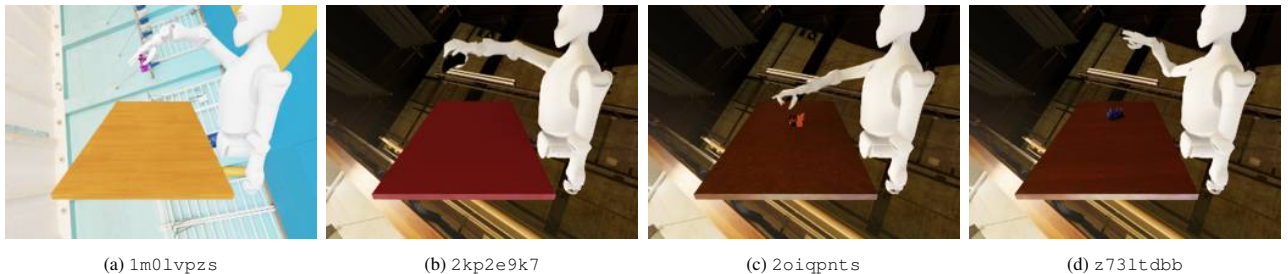


Figure 1: Representative left-camera RGB inputs embedded in `step_000002000_attempt_0001_success.html`. Samples (a)–(c) were selected for high teacher–student disagreement, with ℓ_2 values 9.7506, 17.0997, and 51.9710, respectively; (d) was selected for low predicted success (0.2734). All four records have `unsafe=true`. The transmitted images are 256×192 JPEGs (quality 70). This archived call tagged the images as `detail="low"`; new calls transmit the same image format with `detail="high"`.

The following is the full prompt template used by the runtime implementation. Braced terms denote values serialized at query time, and the visual-sample block is repeated once for each attached image.

VLM Prompt Example

System message:

You tune safety thresholds and return compact JSON only.

User message:

You are a conservative threshold advisor for DexSafeDagger robot distillation. Do not decide intervention directly. Recommend thresholds for the existing gates.

Task: Tune teacher-student disagreement and predicted-success thresholds for dexterous robot distillation.

Existing arbitration:

- intervene if $\text{teacher_student_l2} \geq \text{l2_threshold}$
- or if $\text{predictor_success} \leq \text{success_threshold}$

Return only strict JSON with keys: l2_threshold , success_threshold , confidence , reason . Keep changes conservative. Prefer small threshold shifts unless the recent window shows clear over- or under-intervention.

Base thresholds and clamps:

- $\text{base_l2_threshold} = \{\text{base_l2_threshold}\}$
- $\text{allowed_l2_range} = \{\text{l2_min}, \text{l2_max}\}$
- $\text{base_success_threshold} = \{\text{base_success_threshold}\}$
- $\text{allowed_success_range} = \{\text{success_min}, \text{success_max}\}$

Current advisor thresholds:

- $\text{current_l2_threshold} = \{\text{current_l2_threshold}\}$
- $\text{current_success_threshold} = \{\text{current_success_threshold}\}$

Recent rollout statistics:

```
{
  "frame": {frame}, "intervention_rate": {beta},
  "step": {step},
  "sample_count": {N}, "l2_threshold": {tau_delta},
  "success_threshold": {tau_Q}, "unsafe_episode_rate":
  {r_unsafe},
  "l2": {"min": {v}, "max": {v}, "mean": {v},
    "p10": {v}, "p50": {v}, "p90": {v}},
  "success": {"min": {v}, "max": {v}, "mean": {v},
    "p10": {v}, "p50": {v}, "p90": {v}},
  "unsafe_reason_prop": { "hand_too_far": {v},
    "harmful_collision": {v}, "object_out_of_bound":
    {v},
    "palm_flipped": {v}},
  "visual_buffer_size": {B}, "visual_samples_attached":
  {M},
  "visual_samples": [{ "env_id": {v}, "frame": {v},
    "image_attached": true, "image_key": {v},
    "source": {high_l2 | low_success | unsafe_triggered},
    "l2_threshold": {v}, "success_threshold": {v},
    "teacher_student_l2": {v}, "predictor_success":
    {v},
    "object_id": {v}, "object_name": {v},
    "object_env_count": {v},
    "step": {v}, "unsafe": {true | false}}, ...]
}
```

For each selected visual sample $i=1, \dots, M$:
 Visual sample {i} metadata: {the corresponding record above, excluding "image_attached"}
 <IMAGE {i}: JPEG data URL, detail="high">

The earlier abbreviated prompt omitted the distribution summaries, threshold clamps, unsafe-reason breakdown, visual-buffer context, and per-image metadata; it also used the nonexistent output key `disagreement_threshold` and omitted `confidence`. The runtime prompt uses `l2_threshold`. For reproducibility, the template above preserves the literal inclusive comparison symbols sent to the VLM. The executed controller uses the strict tests in the intervention equation above; the distinction affects only exact equality at a threshold and does not change the VLM's role as a threshold advisor.

B.2 Model and Input Configuration

We used the OpenAI GPT-5.5 multimodal model through the Chat Completions endpoint (`/v1/chat/completions`, `model="gpt-5.5"`). Each request set `stream=false` and `max_completion_tokens=1024`; no temperature or reasoning-effort parameter was sent, so the provider defaults applied. The model received the textual rollout statistics above and up to eight RGB images, encoded as downsampled JPEG data URLs with high visual detail. We issued the first VLM query after 2,000 online distillation steps and subsequently queried it every 10,000 steps, resulting in 10 calls over the 100,000-step run. The archived API reports, which were generated with the previous low-detail image setting, record 30,641 prompt tokens and 4,841 completion tokens, or 35,482 tokens in total.

B.3 Output Processing and Threshold Updates

The requested machine-readable output is a top-level JSON object with the following schema:

```
{
  "l2_threshold": <number>, "success_threshold": <number>,
  "confidence": <number>, "reason": <string>
}
```

The two returned thresholds are absolute candidates. Each candidate is clipped to a fixed range around its initial value and then exponentially smoothed with the previously applied threshold. For $x \in \{\delta, Q\}$,

$$\bar{\tau}_x^{(k)} = \text{clip}\left(\hat{\tau}_x^{(k)}, c_x^{\min} \tau_x^{(0)}, c_x^{\max} \tau_x^{(0)}\right), \quad \tau_x^{(k+1)} = (1 - \alpha) \tau_x^{(k)} + \alpha \bar{\tau}_x^{(k)}.$$

We use $\alpha = 0.1$ and $(c_x^{\min}, c_x^{\max}) = (0.5, 1.5)$ for both thresholds. The updated values are applied from the next control step through the intervention gate $\|a_t^S - a_t^T\|_2 > \tau_\delta \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q$; lowering τ_δ or raising τ_Q increases teacher intervention.

The `confidence` and `reason` fields are logged for diagnostics and do not alter the update. This adaptation is used only during simulation distillation; physical deployment uses the frozen RGB student alone.

If a VLM request fails or its output cannot be parsed, the controller retains the previously applied thresholds and tries again at the next scheduled update. None of the reported trials encountered a condition that activated this fallback.

C Additional Experimental Results

C.1 Bellman Critic Evaluation

We conduct an independent evaluation of the Bellman critic under the rollout setting of ablation (b), in which the student policy remains in control for the entire episode and the teacher never intervenes. The critic runs in parallel as a passive evaluator: it predicts the safe-lift-success value of each proposed student state–action pair, but its prediction does not alter the executed action or trigger teacher takeover. Consequently, unsafe terminal events remain observable, allowing us to measure whether the critic signaled low predicted safe-lift success before each observed unsafe termination.

An unsafe episode is counted as detected if the predicted safe-lift-success value falls below the deployed threshold at least once before the unsafe termination; otherwise, it is counted as a false negative. A safe-success episode is counted as a false alarm if the critic falls below the threshold at least once. For detected unsafe episodes, warning time is measured from the first threshold crossing to the unsafe termination.

Table 3: Independent evaluation of the Bellman critic under the student-controlled rollout setting of ablation (b).

Metric	Result
Evaluation episodes	$[N]$
Unsafe episodes	$[N_u]$
Unsafe-event detection rate	$[x\%]$
Unsafe false-negative rate	$[x\%]$
Safe-success false-alarm rate	$[x\%]$
Median warning time	$[xs]$

C.2 Operating-Point Analysis of the Bellman Critic

The Bellman critic predicts the joint acceptable outcome of successful lifting without an unsafe event. To make the behavior of this joint target transparent, we partition the independently evaluated episodes into three mutually exclusive outcomes: safe lift success, safe lift failure, and unsafe termination. A safe lift failure reaches the episode horizon without satisfying the lift criterion and without triggering any of the four unsafe termination conditions. For each outcome, we report how often the critic falls below the deployed intervention threshold before the episode ends.

Table 4: Bellman-critic warnings stratified by episode outcome under the student-controlled rollout setting of ablation (b).

Episode outcome	Episodes	Critic warned	Warning rate
Safe lift success	$[N_{SS}]$	$[TBD]$	$[r_{SS}\%]$
Safe lift failure	$[N_{SF}]$	$[TBD]$	$[r_{SF}\%]$
Unsafe termination	$[N_U]$	$[TBD]$	$[r_U\%]$

Here, r_U is the unsafe-event detection rate, $1 - r_U$ is the unsafe false-negative rate, and r_{SS} is the safe-success false-alarm rate. Because the critic is trained on the joint safe-lift-success target, r_{SF} measures how often it rejects a safe but unsuccessful trajectory and is not treated as a safety false alarm.

C.3 Ablations Across Random Seeds

All ablations use the same training and computational resource budget and differ only in the intervention mechanism. We evaluate each variant using five random seeds, $\{40, 41, 42, 43, 44\}$, and report the mean $\pm$ standard deviation across seeds.

Table 5: Ablation results across five random seeds $\{40, 41, 42, 43, 44\}$. Lift success and unsafe rate are reported as mean $\pm$ standard deviation across seeds.

Metric	Ablation (a)	Ablation (b)	Ablation (c)	Ablation (d)	Ablation (e)	Ablation (f)	Ablation (g)	Ablation (h)
Lift Success (%) $\uparrow$	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD
Unsafe Rate (%) $\downarrow$	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD	TBD $\pm$ TBD

C.4 Complete Real-World Evaluation and Demonstrations



Figure 2: **Sixteen-object demonstration inventory:** (1) Horse, (2) Teapot, (3) Sphinx, (4) Dragon, (5) Homer bust, (6) Ogre, (7) Cheburashka, (8) Lucy statue, (9) Beethoven bust, (10) Igea bust, (11) Dog pair, (12) Spot dog, (13) Monkey head, (14) Blub character, (15) Statue, and (16) Dinosaur. Each numbered panel is a placeholder for the corresponding object demonstration.

We will report results for the remaining 16 real-world test objects that are not shown in the manuscript’s six-object table, using the same success and safety breakdown. The two eight-object blocks below separate task success from the observed unsafe failure modes. The entries below are provisional synthetic values for layout checking and will be replaced with the measured results.

Table 6: Provisional per-object results for the remaining 16 objects (30 episodes per object). Synthetic values are shown for layout checking only; **OOD objects** are shaded.

Metric (%)	Horse	Teapot	Sphinx	Dragon	Homer	Ogre	Cheburashka	Lucy
Lift Success	83.3	90.0	80.0	86.7	76.7	83.3	90.0	80.0
Unsafe Rate Avg	16.7	13.3	23.3	20.0	26.7	16.7	10.0	23.3
Harmful collision	3.3	0.0	5.0	6.7	13.3	3.3	3.3	6.7
Object escape	13.3	6.7	11.0	13.3	10.0	13.3	6.7	13.3
Palm flipped	0.0	0.0	0.0	0.0	3.3	0.0	0.0	3.3
Hand too far	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Metric (%)	Beethoven	Igea	Dog Pair	Spot	Monkey head	Blub	Statue	Dinosaur
Lift Success	86.7	73.3	70.0	63.3	73.3	60.0	66.7	56.7
Unsafe Rate Avg	20.0	30.0	33.3	40.0	30.0	43.3	36.7	46.7
Harmful collision	6.7	13.3	16.7	20.0	13.3	23.3	16.7	26.7
Object escape	13.3	13.3	13.3	16.7	13.3	16.7	16.7	16.7
Palm flipped	0.0	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Hand too far	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

[Report 95% confidence intervals and statistical comparisons.]

C.5 Physical Baseline Comparison

We deploy the frozen RGB student policies resulting from ablations (d) and (f) on the real robot. Ablation (d) uses the DexTraH-RGB teacher-training recipe and the stationary DexSafeDagger intervention policy, whereas ablation (f) is the complete VLM-DexSafeDagger pipeline. In both cases, only the resulting student policy is deployed; the teacher, Bellman critic, VLM, and threshold-adaptation mechanism are not active during physical evaluation. We evaluate both policies using the same objects, initial-condition protocol, and number of trials. We report lift-success and unsafe-event rates with 95% confidence intervals, together with the corresponding p -values for the between-policy comparisons.

Table 7: Overall real-world comparison of the frozen student policies resulting from ablations (d) and (f), pooled across all 22 objects. Rates are reported with 95% confidence intervals.

Metric	Ablation (d)	Ablation (f)	p -value
Lift success (%)	[TBD (95% CI: TBD-TBD)]	[TBD (95% CI: TBD-TBD)]	[TBD]
Unsafe-event rate (%)	[TBD (95% CI: TBD-TBD)]	[TBD (95% CI: TBD-TBD)]	[TBD]